

Qualification Pack



EV Charging Installation Technician

QP Code: SGJ/Q4011

Version: 1.0

NSQF Level: 4.5

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Qualification Pack

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SGJ/Q4011: EV Charging Installation Technician

Brief Job Description

The EV Charging Installation Technician qualification provides learners with the knowledge and skills to manage the complete lifecycle of EV charging infrastructure-covering planning, installation, operation, and maintenance. The program includes training in EV science, relevant policies, digital tools like software-based charging and GPS mapping, as well as site readiness and safety protocols. Participants will be equipped to lead teams, ensure efficient station operations, and support the growth of India EV ecosystem. This Qualification will support EV industries, by providing skilled workers for management of EV charging infrastructure as Charging Station Supervisors. EV industry has rolled down several EV models in 2-Wheeler, 3-Wheeler and 4-Wheeler segments and have already put significant charging infrastructure in many Indian cities.

Personal Attributes

An EV Charging Installation Technician should be technically skilled, safety-conscious, and detail-oriented. They must possess leadership qualities, be adaptable to digital tools, and committed to supporting sustainable EV infrastructure.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SGJ/N4079: Introduce Basics of EV Science and Impact on Low Carbon Transport.](#)
2. [SGJ/N4080: Demonstrate EV Charging through Computer/Mobile Applications](#)
3. [SGJ/N4081: Introduce Policy and Regulatory Ecosystem for Electric Vehicle Charging Stations](#)
4. [SGJ/N4082: Perform Pre-Installation Planning and Site Readiness for EV Charging Infrastructure](#)
5. [SGJ/N4083: Perform Erection and Installation of EV Charging Stations](#)
6. [SGJ/N4084: Conduct Testing and Commissioning of EV Charging Stations](#)
7. [SGJ/N4085: Introduce GPS Mapping and Explain Its Importance for EV Charging](#)
8. [SGJ/N4086: Operate and Maintain EV Charging Station Efficiently](#)
9. [SGJ/N4087: Follow Basic Health and Safety Practices in EV Charging Station Operations](#)
10. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Qualification Pack

Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Technician
Country	India
NSQF Level	4.5
Credits	15
Aligned to NCO/ISCO/ISIC Code	NCO-2015/ 3123.0400
Minimum Educational Qualification & Experience	Completed 1st year of UG (UG Certificate) with 1 Year of experience OR 12th grade Pass (SCIENCE STREAM) with 1.5 years of experience OR 10th grade pass with 3 Years of experience electrical, mechanical, I&C, power sector OR Previous relevant Qualification of NSQF Level (3.5) with 1.5 years of experience relevant experience in electrical, mechanical, I&C, power sector OR Previous relevant Qualification of NSQF Level (3.0) with 3 Years of experience relevant experience in electrical, mechanical, I&C, power sector
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	07/05/2028
NSQC Approval Date	08/05/2025
Version	1.0
Reference code on NQR	QG-4.5-ES-04259-2025-V1-SCGJ
NQR Version	1

Remarks:



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Total 450 hours (Theory: 240 hours with employability skills, Practical: 150 hours, OJT: 60 hours)

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SGJ/N4079: Introduce Basics of EV Science and Impact on Low Carbon Transport.

Description

This unit outlines the foundational knowledge required to understand electric vehicle (EV) technology and its role in promoting low-carbon transportation. It includes the basic principles of EV science, components and functioning of electric vehicles, and the environmental benefits of EV adoption.

Scope

The scope covers the following :

- The Scope covers the foundational understanding of electric vehicle (EV) technology and its relevance to low-carbon transport systems. It includes the basic science behind EVs, key components and working principles, environmental and economic advantages, and the role of EVs in reducing emissions and supporting clean and sustainable mobility solutions.

Elements and Performance Criteria

Basic EV science and Impact of EV to build low carbon transportation; Comparison between ICE and Electric Vehicles

To be competent, the user/individual on the job must be able to:

- PC1.** Understand the key components of an electric vehicle
- PC2.** Comprehend the types of batteries used in EVs, their characteristics, and their role in energy storage and delivery.
- PC3.** Explore the environmental benefits of EVs
- PC4.** Understand the concept of life cycle analysis and how it applies to EVs, comparing it with the life cycle of ICE vehicles.
- PC5.** Evaluate the total cost of ownership for EVs, including factors like purchase price, fuel/electricity costs, maintenance, and government incentives.
- PC6.** Explore the growth and challenges of EV charging infrastructure, including the importance of public/home charging stations.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Basic principles of electric vehicle (EV) technology.
- KU2.** Components of EVs including motors, batteries, and controllers.
- KU3.** Types and characteristics of EV batteries (e.g., Li-ion, Lead-acid).
- KU4.** Total cost of ownership comparison between EV and ICE vehicles.
- KU5.** Charging systems and power requirements for EVs.
- KU6.** Role of EVs in achieving sustainable transportation goals.
- KU7.** National and global trends in EV adoption.

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Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Read and interpret technical diagrams and battery specifications.
- GS2.** Communicate EV benefits to customers and stakeholders.
- GS3.** Use digital tools to research EV trends and technologies.
- GS4.** Analyze cost and efficiency data using basic software tools.
- GS5.** Document findings on EV vs. ICE comparisons.
- GS6.** Apply basic mathematical calculations (e.g., power, energy).
- GS7.** Engage in group discussions on climate change and transport.
- GS8.** Follow sustainability practices in professional conduct.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Basic EV science and Impact of EV to build low carbon transportation; Comparison between ICE and Electric Vehicles</i>	50	-	-	-
PC1. Understand the key components of an electric vehicle	6	-	-	-
PC2. Comprehend the types of batteries used in EVs, their characteristics, and their role in energy storage and delivery.	8	-	-	-
PC3. Explore the environmental benefits of EVs	8	-	-	-
PC4. Understand the concept of life cycle analysis and how it applies to EVs, comparing it with the life cycle of ICE vehicles.	8	-	-	-
PC5. Evaluate the total cost of ownership for EVs, including factors like purchase price, fuel/electricity costs, maintenance, and government incentives.	10	-	-	-
PC6. Explore the growth and challenges of EV charging infrastructure, including the importance of public/home charging stations.	10	-	-	-
NOS Total	50	-	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4079
NOS Name	Introduce Basics of EV Science and Impact on Low Carbon Transport.
Sector	Green Jobs
Sub-Sector	
Occupation	Technician
NSQF Level	4.5
Credits	1.0
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

SGJ/N4080: Demonstrate EV Charging through Computer/Mobile Applications

Description

This unit outlines the skills and knowledge required to operate electric vehicle (EV) charging stations using computer and mobile applications. It includes understanding the components and user interface of software-driven EV charging systems, managing charging sessions, monitoring real-time data, processing payments, and troubleshooting common issues.

Scope

The scope covers the following :

- The scope covers the use of digital platforms computer and mobile applications for operating and managing EV charging stations. It includes initiating and monitoring charging sessions, handling user authentication and payments, analyzing charging data, and resolving common software-related issues. The scope also involves ensuring secure and user-friendly interactions to enhance the efficiency and accessibility of EV charging infrastructure.

Elements and Performance Criteria

Demo of computer/mobile use for software driven EV charging (Master Class)

To be competent, the user/individual on the job must be able to:

- PC1.** Understand the essentials of document creation, formatting, and editing in Word, including templates, styles, and collaboration tools.
- PC2.** Learn spreadsheet management, data entry, functions, formulas, and data analysis using Excel, including charts and graphs.
- PC3.** Gain insight into the components and architecture of software-driven EV charging systems, including charging stations, charging management software, and communication protocols.
- PC4.** Navigate the user interface of software-driven EV charging systems, monitor charging progress, and troubleshooting common issues.
- PC5.** Explore data collection and management in EV charging, including billing, usage tracking, and remote monitoring capabilities.
- PC6.** Successfully initiate and manage EV charging sessions using software-driven systems, including user authentication and payment processing.
- PC7.** Identify and resolve common issues that may arise during EV charging
- PC8.** Use software-driven tools to monitor and collect charging data, including billing and usage statistics.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Basics of computer and mobile operations.
- KU2.** Overview of EV charging software platforms.

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- KU3.** Functionalities of EV charging apps and dashboards.
- KU4.** Steps to initiate, monitor, and stop charging sessions.
- KU5.** Role of software in managing billing and user authentication.
- KU6.** Troubleshooting steps for software-related issues.
- KU7.** Importance of secure data handling and privacy.
- KU8.** Integration of software with charging station hardware.
- KU9.** Use of data logs and reports for analysis.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Operate mobile apps and PC-based EV charging software.
- GS2.** Communicate app usage steps to users.
- GS3.** Record and store charging session data securely.
- GS4.** Respond to app-based troubleshooting queries.
- GS5.** Perform basic maintenance of software interfaces.
- GS6.** Collaborate with technical support for issue resolution.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Demo of computer/mobile use for software driven EV charging (Master Class)</i>	20	30	-	-
PC1. Understand the essentials of document creation, formatting, and editing in Word, including templates, styles, and collaboration tools.	4	-	-	-
PC2. Learn spreadsheet management, data entry, functions, formulas, and data analysis using Excel, including charts and graphs.	4	-	-	-
PC3. Gain insight into the components and architecture of software-driven EV charging systems, including charging stations, charging management software, and communication protocols.	4	-	-	-
PC4. Navigate the user interface of software-driven EV charging systems, monitor charging progress, and troubleshooting common issues.	4	-	-	-
PC5. Explore data collection and management in EV charging, including billing, usage tracking, and remote monitoring capabilities.	4	-	-	-
PC6. Successfully initiate and manage EV charging sessions using software-driven systems, including user authentication and payment processing.	-	10	-	-
PC7. Identify and resolve common issues that may arise during EV charging	-	10	-	-
PC8. Use software-driven tools to monitor and collect charging data, including billing and usage statistics.	-	10	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4080
NOS Name	Demonstrate EV Charging through Computer/Mobile Applications
Sector	Green Jobs
Sub-Sector	
Occupation	Technician
NSQF Level	4.5
Credits	1.0
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

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SGJ/N4081: Introduce Policy and Regulatory Ecosystem for Electric Vehicle Charging Stations

Description

Gain comprehensive knowledge of the policy and regulatory ecosystem for EV charging stations, including safety standards, permitting, land use, environmental considerations, government incentives, and international best practices.

Scope

The scope covers the following :

- Covers the legal, procedural, and environmental aspects of planning, establishing, and operating EV charging stations, along with incentives and global regulatory insights.

Elements and Performance Criteria

Policy & Regulatory Ecosystem of Electric Vehicle (EV) Charging Stations

To be competent, the user/individual on the job must be able to:

- PC1.** Understand the existing policy landscape for EV charging stations, covering areas such as safety standards, permitting processes, land use regulations, and grid integration.
- PC2.** Gain insight into the regulatory framework that governs the siting, design, and operation of EV charging stations, including legal requirements and codes of practice.
- PC3.** Explore government incentives, subsidies, and tax benefits that promote the development of EV charging infrastructure and encourage EV adoption.
- PC4.** Learn about environmental regulations and considerations, including emissions reductions and sustainability requirements, related to EV charging infrastructure
- PC5.** Navigate the permitting and zoning requirements for EV charging stations, including land use permissions and location-specific regulations.
- PC6.** Explore international best practices in EV charging station regulation and policy, enabling you to adapt and improve local policies and standards.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** EV charging station policies, safety standards, and grid integration.
- KU2.** Legal requirements and codes for siting, design, and operation.
- KU3.** Government incentives, subsidies, and tax benefits.
- KU4.** Environmental and sustainability regulations.
- KU5.** Permitting, zoning, and land use rules.
- KU6.** International best practices and regulatory models.

Generic Skills (GS)

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User/individual on the job needs to know how to:

- GS1.** Ability to interpret and apply EV charging-related policies, standards, and regulations.
- GS2.** Skill in identifying and leveraging government incentives and subsidies for EV infrastructure.
- GS3.** Competence in navigating permitting, zoning, and land use procedures.
- GS4.** Ability to assess environmental and sustainability compliance requirements.
- GS5.** Research and analytical skills to compare and adapt international best practices.
- GS6.** Effective communication and coordination with regulatory authorities and stakeholders.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Policy & Regulatory Ecosystem of Electric Vehicle (EV) Charging Stations</i>	50	-	-	-
PC1. Understand the existing policy landscape for EV charging stations, covering areas such as safety standards, permitting processes, land use regulations, and grid integration.	10	-	-	-
PC2. Gain insight into the regulatory framework that governs the siting, design, and operation of EV charging stations, including legal requirements and codes of practice.	10	-	-	-
PC3. Explore government incentives, subsidies, and tax benefits that promote the development of EV charging infrastructure and encourage EV adoption.	7	-	-	-
PC4. Learn about environmental regulations and considerations, including emissions reductions and sustainability requirements, related to EV charging infrastructure	6	-	-	-
PC5. Navigate the permitting and zoning requirements for EV charging stations, including land use permissions and location-specific regulations.	7	-	-	-
PC6. Explore international best practices in EV charging station regulation and policy, enabling you to adapt and improve local policies and standards.	10	-	-	-
NOS Total	50	-	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4081
NOS Name	Introduce Policy and Regulatory Ecosystem for Electric Vehicle Charging Stations
Sector	Green Jobs
Sub-Sector	
Occupation	Technician
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

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SGJ/N4082: Perform Pre-Installation Planning and Site Readiness for EV Charging Infrastructure

Description

It includes site assessment, load analysis, equipment selection, safety protocols, and environmental considerations for deploying reliable and sustainable EV charging infrastructure.

Scope

The scope covers the following :

- This covers the installation, commissioning, and maintenance of EV charging stations in compliance with local codes, ensuring safety, efficiency, and scalability.

Elements and Performance Criteria

Planning and Preparedness for installation and Commissioning process of EV charging station

To be competent, the user/individual on the job must be able to:

- PC1.** Master regulations for EV charging installations, ensuring compliance and safe deployment within local codes and standards.
- PC2.** Learn load analysis techniques, assess grid impact, and propose upgrades, optimizing electrical infrastructure for EV stations.
- PC3.** Grasp principles of sizing infrastructure to support diverse charging stations, ensuring scalability and efficient operation.
- PC4.** Develop safety protocols and emergency plans, prioritizing user and personnel safety in EV charging station environments.
- PC5.** Assess sites for EV charging, considering accessibility and power; tailor installations to user demand and location suitability.
- PC6.** Master equipment selection aligned with station needs and budget, ensuring optimal charging infrastructure.
- PC7.** Develop skills in safe electrical installation, covering components, charging stations, and associated equipment
- PC8.** Learn commissioning processes, including testing, software configuration, and safety checks, ensuring station functionality and reliability.
- PC9.** Understanding the environmental impact of EV charging stations and exploring strategies for reducing energy consumption and environmental footprint.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Knowledge of national and local electrical codes, safety standards, and permitting processes for EV charging station installations.
- KU2.** Understanding of load analysis methods, power demand forecasting, and evaluating grid capacity to recommend infrastructure upgrades.

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- KU3.** Familiarity with principles of designing and sizing electrical systems to support different types and scales of EV chargers.
- KU4.** Ability to assess site conditions, including power availability, accessibility, and user demand, for optimal charger placement.
- KU5.** Understanding of electrical safety, risk mitigation strategies, and emergency response procedures in EV charging environments.
- KU6.** Awareness of the environmental impact of EV charging stations and strategies to minimize energy consumption and carbon footprint.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to perform load analysis, assess grid impact, and interpret technical data for infrastructure planning and upgrades.
- GS2.** Skilled in selecting, installing, and configuring EV charging equipment and associated electrical components safely and accurately.
- GS3.** Competence in identifying site-specific challenges and proposing tailored, efficient, and scalable EV charging solutions.
- GS4.** Strong understanding of safety standards, local codes, and the ability to develop and implement effective safety and emergency protocols.
- GS5.** Awareness of sustainable practices and ability to incorporate energy-efficient strategies to reduce environmental impact.
- GS6.** Ability to work effectively with stakeholders, including utility providers, site owners, and team members, to ensure smooth project execution.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Planning and Preparedness for installation and Commissioning process of EV charging station</i>	23	27	-	-
PC1. Master regulations for EV charging installations, ensuring compliance and safe deployment within local codes and standards.	8	-	-	-
PC2. Learn load analysis techniques, assess grid impact, and propose upgrades, optimizing electrical infrastructure for EV stations.	6	-	-	-
PC3. Grasp principles of sizing infrastructure to support diverse charging stations, ensuring scalability and efficient operation.	4	-	-	-
PC4. Develop safety protocols and emergency plans, prioritizing user and personnel safety in EV charging station environments.	5	-	-	-
PC5. Assess sites for EV charging, considering accessibility and power; tailor installations to user demand and location suitability.	-	5	-	-
PC6. Master equipment selection aligned with station needs and budget, ensuring optimal charging infrastructure.	-	4	-	-
PC7. Develop skills in safe electrical installation, covering components, charging stations, and associated equipment	-	6	-	-
PC8. Learn commissioning processes, including testing, software configuration, and safety checks, ensuring station functionality and reliability.	-	6	-	-
PC9. Understanding the environmental impact of EV charging stations and exploring strategies for reducing energy consumption and environmental footprint.	-	6	-	-
NOS Total	23	27	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4082
NOS Name	Perform Pre-Installation Planning and Site Readiness for EV Charging Infrastructure
Sector	Green Jobs
Sub-Sector	
Occupation	Technician
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

SGJ/N4083: Perform Erection and Installation of EV Charging Stations

Description

Outlines the skills required to safely and efficiently install EV charging stations while ensuring adherence to technical and regulatory standards.

Scope

The scope covers the following :

- Covers interpretation of technical specifications, compliance evaluation, physical installation, and quality checks for EV charging station components.

Elements and Performance Criteria

Erection and Installation of EV Charging Station

To be competent, the user/individual on the job must be able to:

- PC1.** Interpret technical specifications for charging station components.
- PC2.** Evaluate the implications of non-compliance.
- PC3.** Demonstrate proficiency in the physical installation process.
- PC4.** Conduct quality checks during and after installation.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Basic electrical principles and technical specifications related to EV charging station components (e.g., connectors, power units, safety systems).
- KU2.** Applicable industry standards, regulations, and manufacturer guidelines for the installation of EV charging infrastructure.
- KU3.** Potential risks and consequences of non-compliance with safety, electrical, and regulatory norms.
- KU4.** Tools, equipment, and materials required for the physical installation of charging stations.
- KU5.** Procedures for conducting pre- and post-installation quality checks to ensure functionality and safety.
- KU6.** Methods for identifying faults, reporting deviations, and performing corrective actions during installation.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to read and interpret technical manuals, wiring diagrams, and installation guides related to EV charging systems.

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- GS2.** Analytical thinking to assess technical requirements and evaluate the risks associated with non-compliance.
- GS3.** Manual dexterity and physical coordination to carry out precise installation tasks using appropriate tools and equipment.
- GS4.** Attention to detail to ensure accurate installation and adherence to quality and safety standards.
- GS5.** Problem-solving skills to identify, troubleshoot, and rectify issues during installation and quality checks.
- GS6.** Communication skills to report technical issues, coordinate with team members, and document installation processes.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Erection and Installation of EV Charging Station</i>	20	30	-	-
PC1. Interpret technical specifications for charging station components.	10	-	-	-
PC2. Evaluate the implications of non-compliance.	10	-	-	-
PC3. Demonstrate proficiency in the physical installation process.	-	15	-	-
PC4. Conduct quality checks during and after installation.	-	15	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4083
NOS Name	Perform Erection and Installation of EV Charging Stations
Sector	Green Jobs
Sub-Sector	
Occupation	Technician
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

SGJ/N4084: Conduct Testing and Commissioning of EV Charging Stations

Description

The individual will be responsible for identifying testing requirements, developing and executing a commissioning plan, performing accurate testing procedures, and demonstrating proficiency in the end-to-end commissioning of EV charging stations.

Scope

The scope covers the following :

- It covers the skills and knowledge required to test, commission, and ensure the proper functioning of various components of Electric Vehicle (EV) charging infrastructure.

Elements and Performance Criteria

Testing and Commissioning of EV Charging Station

To be competent, the user/individual on the job must be able to:

- PC1.** Identify testing requirements for various charging station components.
- PC2.** Develop a commissioning plan for EV charging infrastructure.
- PC3.** Execute testing procedures accurately.
- PC4.** Demonstrate proficiency in the commissioning process.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understanding the types, architecture, and components of EV charging stations (AC/DC chargers, connectors, power modules, communication interfaces).
- KU2.** Knowledge of relevant national and international testing standards (e.g., IEC, IS, IEEE) applicable to EVSE (Electric Vehicle Supply Equipment).
- KU3.** Familiarity with steps involved in planning, executing, and validating the commissioning of EV charging infrastructure.
- KU4.** Understanding the use of testing tools like multimeters, insulation testers, power analyzers, and software diagnostic tools.
- KU5.** Awareness of electrical safety procedures, grounding methods, and regulatory compliance during testing and commissioning.
- KU6.** Knowledge of preparing test reports, commissioning checklists, and maintaining accurate records for audits and performance tracking.

Generic Skills (GS)

User/individual on the job needs to know how to:

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- GS1.** Ability to assess technical specifications and determine appropriate testing requirements for various EV charging components.
- GS2.** Proficiency in developing and implementing a structured commissioning plan for EV charging infrastructure
- GS3.** Accuracy in executing testing procedures and identifying faults or deviations in system performance.
- GS4.** Capability to troubleshoot issues during testing and commissioning with minimal supervision.
- GS5.** Effective communication of test results, issues, and recommendations through clear documentation and verbal reporting.
- GS6.** Ability to work efficiently as part of a team, coordinating with engineers, electricians, and project managers during the commissioning process.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Testing and Commissioning of EV Charging Station</i>	30	20	-	-
PC1. Identify testing requirements for various charging station components.	15	-	-	-
PC2. Develop a commissioning plan for EV charging infrastructure.	15	-	-	-
PC3. Execute testing procedures accurately.	-	10	-	-
PC4. Demonstrate proficiency in the commissioning process.	-	10	-	-
NOS Total	30	20	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4084
NOS Name	Conduct Testing and Commissioning of EV Charging Stations
Sector	Green Jobs
Sub-Sector	
Occupation	Technician
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

SGJ/N4085: Introduce GPS Mapping and Explain Its Importance for EV Charging

Description

It covers the basics of GPS mapping, interpretation of geospatial data, and its application in evaluating and selecting optimal locations for EV charging stations. It includes hands-on training to apply GPS tools in real-world EV infrastructure projects.

Scope

The scope covers the following :

- To equip individuals with fundamental knowledge and practical skills in GPS mapping for effective site selection, navigation, and optimization of Electric Vehicle (EV) charging infrastructure.

Elements and Performance Criteria

GPS mapping and its importance

To be competent, the user/individual on the job must be able to:

- PC1.** Fundamentals of GPS mapping and its importance
- PC2.** Interpret GPS data for site selection and navigation.
- PC3.** Evaluate the role of mapping in optimizing charging station locations.
- PC4.** Hands on training on GPS mapping
- PC5.** Utilize GPS data for real-world applications in EV infrastructure and gathering related information

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Basics of GPS technology and its significance in mapping.
- KU2.** Importance of GPS in EV infrastructure planning.
- KU3.** How to interpret GPS data for site selection and navigation.
- KU4.** Role of mapping in optimizing EV charging station locations.
- KU5.** Hands-on use of GPS tools and software.
- KU6.** Application of GPS data in real-world EV projects.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to operate GPS devices and mapping software effectively.
- GS2.** Skill in analyzing and interpreting geospatial data.



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- GS3.** Decision-making for site selection based on GPS inputs.
- GS4.** Problem-solving in navigation and EV infrastructure planning.
- GS5.** Practical skills in field data collection using GPS tools.
- GS6.** Communication skills to present mapping data and findings clearly.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>GPS mapping and its importance</i>	30	20	-	-
PC1. Fundamentals of GPS mapping and its importance	10	-	-	-
PC2. Interpret GPS data for site selection and navigation.	10	-	-	-
PC3. Evaluate the role of mapping in optimizing charging station locations.	10	-	-	-
PC4. Hands on training on GPS mapping	-	10	-	-
PC5. Utilize GPS data for real-world applications in EV infrastructure and gathering related information	-	10	-	-
NOS Total	30	20	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4085
NOS Name	Introduce GPS Mapping and Explain Its Importance for EV Charging
Sector	Green Jobs
Sub-Sector	
Occupation	Technician
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

SGJ/N4086: Operate and Maintain EV Charging Station Efficiently

Description

The individual will be responsible for implementing operational procedures, preventive maintenance, and troubleshooting tasks, while ensuring a positive user experience and adherence to environmental best practices.

Scope

The scope covers the following :

- It covers the skills and knowledge required to operate, maintain, and troubleshoot EV charging stations, ensuring high functionality, user satisfaction, and minimal environmental impact.

Elements and Performance Criteria

Operating the EV Charging Station

To be competent, the user/individual on the job must be able to:

- PC1.** Develop operational procedures for smooth charging station functionality.
- PC2.** Evaluate user experience and satisfaction.
- PC3.** Demonstrate proficiency in operating station components
- PC4.** Provide excellent customer service.

Maintain the EV Charging Station

To be competent, the user/individual on the job must be able to:

- PC5.** Develop a maintenance schedule and checklist.
- PC6.** Evaluate the environmental impact of maintenance practices.
- PC7.** Implement preventive maintenance strategies.
- PC8.** Demonstrate troubleshooting skills.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Components and functioning of electric vehicle (EV) charging stations
- KU2.** Standard operating procedures for station setup and user interaction
- KU3.** Preventive and corrective maintenance practices for EV infrastructure
- KU4.** Methods to develop and use maintenance schedules and checklists
- KU5.** Principles of customer service and techniques to evaluate user satisfaction
- KU6.** Environmental considerations and impact of maintenance activities

Generic Skills (GS)

User/individual on the job needs to know how to:



Qualification Pack

- GS1.** Communicate effectively with users and team members to ensure smooth operations and customer satisfaction
- GS2.** Follow standard operating procedures and safety protocols at EV charging stations
- GS3.** Use tools, checklists, and digital systems to plan and execute maintenance tasks
- GS4.** Apply analytical thinking to identify, troubleshoot, and resolve technical issues
- GS5.** Manage time efficiently to perform preventive maintenance without disrupting service
- GS6.** Assess and adapt work practices to minimize environmental impact

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Operating the EV Charging Station</i>	20	20	-	-
PC1. Develop operational procedures for smooth charging station functionality.	10	-	-	-
PC2. Evaluate user experience and satisfaction.	10	-	-	-
PC3. Demonstrate proficiency in operating station components	-	10	-	-
PC4. Provide excellent customer service.	-	10	-	-
<i>Maintain the EV Charging Station</i>	40	20	-	-
PC5. Develop a maintenance schedule and checklist.	20	-	-	-
PC6. Evaluate the environmental impact of maintenance practices.	20	-	-	-
PC7. Implement preventive maintenance strategies.	-	10	-	-
PC8. Demonstrate troubleshooting skills.	-	10	-	-
NOS Total	60	40	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4086
NOS Name	Operate and Maintain EV Charging Station Efficiently
Sector	Green Jobs
Sub-Sector	
Occupation	Technician
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

SGJ/N4087: Follow Basic Health and Safety Practices in EV Charging Station Operations

Description

The unit focuses on equipping participants with the skills to apply safety measures in real-world scenarios, including conducting safety drills and simulations to assess effectiveness.

Scope

The scope covers the following :

- It covers the development, communication, implementation, and evaluation of safety protocols to ensure workplace safety and preparedness.

Elements and Performance Criteria

Basic Health and Safety Practices

To be competent, the user/individual on the job must be able to:

- PC1.** Develop and communicate safety protocols.
- PC2.** Evaluate the effectiveness of safety measures
- PC3.** Apply safety measures in practical scenarios
- PC4.** Conduct safety drills and simulations.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Principles and components of effective safety protocols.
- KU2.** Methods for identifying potential hazards and risks.
- KU3.** Techniques for clearly communicating safety procedures to stakeholders.
- KU4.** Tools and criteria for evaluating the effectiveness of safety measures.
- KU5.** Best practices for applying safety measures in various workplace scenarios.
- KU6.** Procedures and guidelines for planning and conducting safety drills and simulations.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to draft and present clear and concise safety protocols.
- GS2.** Effective communication skills for training and briefing teams on safety measures.
- GS3.** Critical thinking to assess risks and implement appropriate safety actions.
- GS4.** Analytical skills to evaluate safety measures and identify areas of improvement.
- GS5.** Team coordination skills for organizing and executing safety drills.



Qualification Pack

GS6. Problem-solving abilities to respond to emergencies and improve safety practices.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Basic Health and Safety Practices</i>	30	20	-	-
PC1. Develop and communicate safety protocols.	15	-	-	-
PC2. Evaluate the effectiveness of safety measures	15	-	-	-
PC3. Apply safety measures in practical scenarios	-	10	-	-
PC4. Conduct safety drills and simulations.	-	10	-	-
NOS Total	30	20	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4087
NOS Name	Follow Basic Health and Safety Practices in EV Charging Station Operations
Sector	Green Jobs
Sub-Sector	
Occupation	Technician
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

Qualification Pack

- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2029
NSQC Clearance Date	06/02/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification.

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Minimum Aggregate Passing % at QP Level : 70

(**Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SGJ/N4079.Introduce Basics of EV Science and Impact on Low Carbon Transport.	50	0	0	0	50	10
SGJ/N4080.Demonstrate EV Charging through Computer/Mobile Applications	20	30	0	0	50	10
SGJ/N4081.Introduce Policy and Regulatory Ecosystem for Electric Vehicle Charging Stations	50	0	0	0	50	10
SGJ/N4082.Perform Pre-Installation Planning and Site Readiness for EV Charging Infrastructure	23	27	0	0	50	10
SGJ/N4083.Perform Erection and Installation of EV Charging Stations	20	30	0	0	50	10
SGJ/N4084.Conduct Testing and Commissioning of EV Charging Stations	30	20	0	0	50	10
SGJ/N4085.Introduce GPS Mapping and Explain Its Importance for EV Charging	30	20	0	0	50	10
SGJ/N4086.Operate and Maintain EV Charging Station Efficiently	60	40	0	0	100	10

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SGJ/N4087.Follow Basic Health and Safety Practices in EV Charging Station Operations	30	20	0	0	50	10
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	10
Total	333	217	-	-	550	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.